

Introduction to Biology

AI Tutor - Biology - Lesson 1

Introduction

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Explanation

Biology is the science of life and living organisms. It studies everything from the tiniest single-celled bacteria to the largest whales, examining how living things grow, reproduce, and interact with their environment. The cell is the basic unit of life: every living organism is made of one or more cells. Groups of similar cells form tissues, tissues form organs, and organs work together as organisms.

Key Points

- The cell is the basic unit of life.
- Tissues are groups of similar cells working together.
- Organs are made of different tissues working together for a function.
- An organism is a complete living thing made of organs working together.

Examples

- A single red blood cell is a cell.
- Muscle tissue is made of many muscle cells together.
- The heart is an organ made of muscle and other tissues.
- A human being is an organism made of many organs.

Summary

Biology studies living organisms, starting from the cell (life's basic unit) up through tissues, organs, and whole organisms.

Practice Questions

- 1 What is the basic unit of life called?
- 2 Put in order from smallest to largest: organ, cell, tissue, organism.
- 3 Name one organ in the human body and its main tissue type.

Quiz

Q1. What is the basic unit of life?

a) Tissue | b) Organ | c) Cell | d) Organism

Q2. A group of similar cells working together is called a:

a) Organ | b) Tissue | c) Organism | d) Molecule

Q3. Which is the correct order from smallest to largest?

a) Organism -> Organ -> Tissue -> Cell | b) Cell -> Tissue -> Organ -> Organism | c) Tissue -> Cell -> Organism -> Organ | d) Organ -> Cell -> Tissue -> Organism